

Problem

Let a, b, c be positive real numbers. Prove that

$$\left(a + \frac{1}{b}\right) \left(b + \frac{1}{c}\right) \left(c + \frac{1}{a}\right) \geq 2(1 + a + b + c).$$

Source: APMO 1998

§1 Normalizing the Product

Introduce the variables

$$x = \frac{1}{ab}, \quad y = \frac{1}{bc}, \quad z = \frac{1}{ca}.$$

Then

$$xyz = \frac{1}{(abc)^2}.$$

Multiplying a, b, c simultaneously by the positive factor abc , we may normalize the inequality so that

$$xyz = 1.$$

Under this substitution,

$$a = \frac{1}{\sqrt{xy}}, \quad b = \frac{1}{\sqrt{yz}}, \quad c = \frac{1}{\sqrt{zx}},$$

and the desired inequality becomes

$$(x + y)(y + z)(z + x) \geq 2(1 + x + y + z).$$

Thus it suffices to prove

$$(x + y)(y + z)(z + x) - 2(1 + x + y + z) \geq 0,$$

subject to

$$xyz = 1.$$

§2 Extracting the Symmetric Structure

Expanding the product,

$$(x + y)(y + z)(z + x) = (x + y + z)(xy + yz + zx) - xyz.$$

Since $xyz = 1$, the inequality is equivalent to

$$(x + y + z)(xy + yz + zx) - 1 \geq 2(1 + x + y + z),$$

or

$$(x + y + z)(xy + yz + zx - 2) \geq 3.$$

Therefore it remains to show

$$(x + y + z)(xy + yz + zx - 2) \geq 3.$$

§3 Harnessing AM–GM

Because $xyz = 1$, the AM–GM inequality yields

$$x + y + z \geq 3.$$

Also,

$$xy + yz + zx \geq 3\sqrt[3]{(xyz)^2} = 3.$$

Hence

$$xy + yz + zx - 2 \geq 1.$$

Combining these estimates,

$$(x + y + z)(xy + yz + zx - 2) \geq 3 \cdot 1 = 3.$$

This is precisely the inequality required above.

§4 The Final Conclusion

We have proved that

$$(x + y)(y + z)(z + x) \geq 2(1 + x + y + z).$$

Reversing the substitution gives

$$\left(a + \frac{1}{b}\right) \left(b + \frac{1}{c}\right) \left(c + \frac{1}{a}\right) \geq 2(1 + a + b + c).$$

Therefore,

$$\boxed{\left(a + \frac{1}{b}\right) \left(b + \frac{1}{c}\right) \left(c + \frac{1}{a}\right) \geq 2(1 + a + b + c)}.$$